

Olanna Health

Encryption & Data Protection Documentation

Version 1.0.0 | Bundle ID: com.olannahealth.app | Effective: 1 March 2026

1. Encryption Declaration Summary

Olanna Health is a women's health and wellness tracking application. This document describes all encryption used within the application for the purposes of Apple App Store compliance (Export Compliance) and transparency.

Declaration Item	Value
App Name	Olanna Health
Bundle Identifier	com.olannahealth.app
Platform	iOS / Android (React Native / Expo)
ITSApplUsesNonExemptEncryption	false
Uses proprietary/custom encryption	No
Contains encryption not provided by the OS	No
Export Compliance Required	Exempt

2. Encryption Technologies Used

2.1 HTTPS / TLS (Transport Layer Security)

All network communications between the Olanna Health mobile client and the backend server are conducted exclusively over **HTTPS** using TLS 1.2 or higher. This encryption is provided by the operating system's built-in networking stack (NSURLSession on iOS, OkHttp on Android) and is not a custom implementation.

- All API requests use HTTPS endpoints via the platform's standard `fetch()` API
- No custom TLS certificates or certificate pinning is implemented
- HTTPS encryption is handled entirely by the OS networking layer

2.2 On-Device Secure Storage (iOS Keychain / Android Keystore)

Sensitive health data (user profile, cycle data, daily symptom logs) is stored using **expo-secure-store**, which leverages:

- **iOS:** Keychain Services with hardware-backed encryption (AES-256-GCM on devices with Secure Enclave)
- **Android:** Android Keystore system with AES encryption
- **Web fallback:** AsyncStorage (non-encrypted local storage, used only for web development)

This encryption is provided entirely by the respective operating system's secure storage APIs. Olanna Health does not implement any custom encryption algorithms for local storage.

2.3 AsyncStorage (Non-Sensitive Data)

Non-sensitive application preferences and UI state are stored using **AsyncStorage**, which uses the platform's standard local storage mechanism. This does not involve encryption.

Data Category	Storage Method	Encryption	Provider
User profile, cycle data, daily logs	expo-secure-store	AES-256 (Keychain / Keystore)	OS-provided
Privacy settings, preferences	AsyncStorage	None (non-sensitive)	OS-provided
Partner access token	expo-secure-store	AES-256 (Keychain / Keystore)	OS-provided
Device identifier	expo-secure-store	AES-256 (Keychain / Keystore)	OS-provided
API communication	HTTPS (TLS 1.2+)	TLS	OS-provided

3. Export Compliance Justification

Olanna Health is **exempt** from export compliance documentation requirements (ECCN 5D002) because:

- **No proprietary encryption:** The app does not contain, implement, or call any custom or third-party encryption libraries beyond what the operating system provides.
- **Standard OS encryption only:** All encryption used (HTTPS/TLS, Keychain, Keystore) is provided by and managed by iOS and Android operating systems.

- **No encryption for obfuscation:** Encryption is not used to obscure app functionality or intellectual property.
- **Exempt use case:** Encryption is used solely for authentication, data protection, and secure communication — all qualifying exemptions under U.S. Bureau of Industry and Security (BIS) Export Administration Regulations (EAR) Category 5, Part 2, Note 4.

The `ITSApplUsesNonExemptEncryption` key in the app's `Info.plist` is set to **false**, confirming this exemption.

4. Data Flow & Encryption Architecture

4.1 Local-First Architecture

Olanna Health follows a **local-first data model**. The majority of user data — including menstrual cycle dates, symptom logs, mood entries, and health profiles — is stored exclusively on the user's device and never transmitted to external servers.

4.2 Server Communication

The only features requiring server communication are:

- **Partner Mode (opt-in):** A limited cycle snapshot is transmitted to the server via HTTPS so a linked partner can view selected data. Access is controlled via partner tokens validated server-side.
- **AI-Powered Features:** When the user interacts with AI health features, queries are sent to the server via HTTPS. The server proxies requests to OpenAI's API. No personal health data is stored server-side beyond the immediate request lifecycle.

4.3 Server-Side Security

- Express.js backend with rate limiting on all API endpoints
- Device-scoped data access via `x-device-id` header
- Partner authentication via secure token validation (`x-partner-token`)
- PostgreSQL database with Drizzle ORM for structured data access
- No plaintext secrets in source code — environment variables used for all credentials

5. Privacy Manifest (iOS)

The app includes an iOS Privacy Manifest (`NSPrivacyAccessedAPITypes`) declaring use of:

API Type	Reason Code	Description
<code>NSPrivacyAccessedAPICategoryUserDefaults</code>	CA92.1	Access preferences and app state via <code>UserDefaults</code> (<code>AsyncStorage</code>)

6. Third-Party SDKs & Encryption

The following third-party components interact with encryption, all using OS-provided mechanisms:

SDK / Library	Purpose	Encryption Usage
expo-secure-store	Secure on-device storage	Uses iOS Keychain / Android Keystore (OS-provided)
React Native fetch API	Network requests	HTTPS/TLS via OS networking stack
@tanstack/react-query	API data fetching/caching	In-memory cache only, no encryption
expo-crypto	Available but used minimally	OS-provided hashing/random generation

None of these SDKs implement custom encryption algorithms. All rely on the host operating system's encryption infrastructure.

7. Compliance Summary

Regulation / Framework	Status
Apple App Store Export Compliance	Exempt — <code>ITSAAppUsesNonExemptEncryption = false</code>
U.S. EAR (Export Administration Regulations)	Exempt — Uses only OS-provided encryption
POPIA (South Africa)	Compliant — Local-first data model, user controls all data
GDPR Principles	Aligned — Data minimisation, purpose limitation, storage limitation

8. Contact Information

For questions regarding encryption, data handling, or compliance:

- **Email:** admin@olanna.health
- **Application:** Olanna Health v1.0.0
- **Developer:** Olanna Health

This document is intended for Apple App Store review and internal compliance records.